



LAB SHEET-04

COST-21053



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PE-2023-027

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Q1)

```
public abstract class StaffMembernew {

    private String fullName;
    private final String staffId;
    protected String department;

    private static int staffCount = 0;

    // Constructor
    public StaffMembernew(String fullName, String staffId, String department) {
        this.fullName = fullName;
        this.staffId = staffId;
        this.department = department;
        staffCount++; /
    }

    // Getters
    public String getFullName() {
        return fullName;
    }

    public String getStaffId() {
        return staffId;
    }
}
```

```
public String getDepartment() {  
    return department;  
}
```

```
// Final method
```

```
public final void displayBasicDetails() {  
    System.out.println("Name: " + fullName);  
    System.out.println("ID: " + staffId);  
    System.out.println("Department: " + department);  
}
```

```
// Abstract method
```

```
public abstract double calculateMonthlyPayment();
```

```
// Static methods
```

```
public static void showSystemName() {  
    System.out.println("=== Staff Management System ===");  
}
```

```
public static int getStaffCount() {  
    return staffCount;  
}
```

```
// Encapsulation methods
```

```
public void changeDepartment(String newDepartment) {  
    if (newDepartment != null && !newDepartment.isEmpty()) {
```

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```
        this.department = newDepartment;
    }
}

public final void showCommonNotice() {
    System.out.println("Notice: All staff must follow university rules.");
}
}
```

Q2) public class Lecturer extends StaffMembernew {

private int courseCount;

private double paymentPerCourse;

public Lecturer(String name, String id, String dept, int courseCount, double paymentPerCourse) {

super(name, id, dept);

this.courseCount = courseCount;

this.paymentPerCourse = paymentPerCourse;

}

@Override

public double calculateMonthlyPayment() {

return courseCount * paymentPerCourse;

}

public void displayLecturerDetails() {

displayBasicDetails();

System.out.println("Courses: " + courseCount);

System.out.println("Payment per Course: " + paymentPerCourse);

}

}

Q3) public class LabAssistant extends StaffMembernew {

private int hoursWorked;

private double hourlyRate;

public LabAssistant(String name, String id, String dept, int hoursWorked, double hourlyRate) {

super(name, id, dept);

this.hoursWorked = hoursWorked;

this.hourlyRate = hourlyRate;

}

@Override

public double calculateMonthlyPayment() {

return hoursWorked * hourlyRate;

}

public void displayLabAssistantDetails() {

displayBasicDetails();

System.out.println("Hours Worked: " + hoursWorked);

System.out.println("Hourly Rate: " + hourlyRate);

}

}

Q4) public final class UniversityPolicy {

public static final String UNIVERSITY_NAME = "ABC University";

public static final double BONUS_RATE = 0.10;

public static void showPolicyHeader() {

System.out.println("=== University Policy ===");

}

public static double calculateBonus(double monthlyPayment) {

return monthlyPayment * BONUS_RATE;

}

}

Q5)

```
public class Lab04Main{

    public static void main(String[] args) {

        // System name
        StaffMembernew.showSystemName();

        // Policy
        UniversityPolicy.showPolicyHeader();
        System.out.println("University: " + UniversityPolicy.UNIVERSITY_NAME);

        // Create objects
        Lecturer lecturer1 = new Lecturer("Ali", "L001", "IT", 3, 5000);
        Lecturer lecturer2 = new Lecturer("Sara", "L002", "CS", 2, 6000);
        LabAssistant assistant = new LabAssistant("John", "A001", "Lab", 120, 50);

        // Change department
        lecturer1.changeDepartment("SE");

        // Display details
        lecturer1.displayLecturerDetails();
        System.out.println("Monthly Payment: " + lecturer1.calculateMonthlyPayment());

        lecturer2.displayLecturerDetails();
        System.out.println("Monthly Payment: " + lecturer2.calculateMonthlyPayment());
    }
}
```



```
assistant.displayLabAssistantDetails();

System.out.println("Monthly Payment: " + assistant.calculateMonthlyPayment());


// Total payment

double total = lecturer1.calculateMonthlyPayment()
    + lecturer2.calculateMonthlyPayment()
    + assistant.calculateMonthlyPayment();


System.out.println("Total Monthly Payment: " + total);


// Staff count

System.out.println("Total Staff: " + StaffMembernew.getStaffCount());


// Common notice

lecturer1.showCommonNotice();

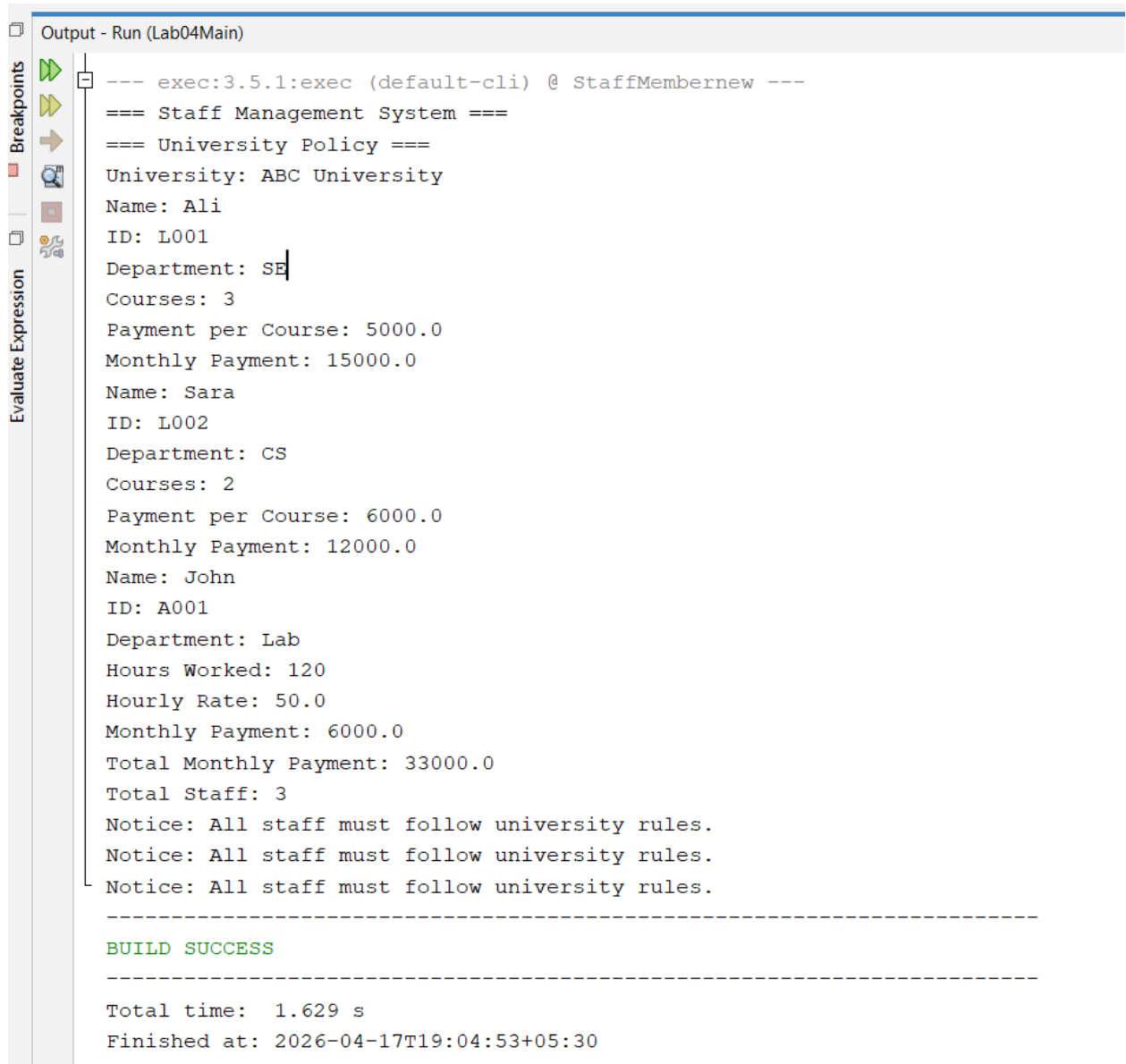
lecturer2.showCommonNotice();

assistant.showCommonNotice();

}

}
```

Out put



```
Output - Run (Lab04Main)

--- exec:3.5.1:exec (default-cli) @ StaffMembernew ---
=== Staff Management System ===
=== University Policy ===
University: ABC University
Name: Ali
ID: L001
Department: SE
Courses: 3
Payment per Course: 5000.0
Monthly Payment: 15000.0
Name: Sara
ID: L002
Department: CS
Courses: 2
Payment per Course: 6000.0
Monthly Payment: 12000.0
Name: John
ID: A001
Department: Lab
Hours Worked: 120
Hourly Rate: 50.0
Monthly Payment: 6000.0
Total Monthly Payment: 33000.0
Total Staff: 3
Notice: All staff must follow university rules.
Notice: All staff must follow university rules.
Notice: All staff must follow university rules.
-----
BUILD SUCCESS
-----

Total time: 1.629 s
Finished at: 2026-04-17T19:04:53+05:30
```